Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 1 – Project Introduction

– Hanoi, Sep 2026 –

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Change Log

Date	A* M, D	In charge	Change Description
20/9/2026	A	Nguyen Hoang Thang	Add I.Overview
20/9/2026	A	Nguyen Hoang Thang	Add II.Product Background
20/9/2026	A	Nguyen Thanh Luan	Add III.Existing Solutions
20/9/2026	A	Nguyen Viet Hoang	Add IV. Proposed Solution
21/9/2026	M	Nguyen Thanh Luan	Add detailed information to IV. Proposed Solution
21/9/2026	A	Nguyen Chi Tung Duong	Add V.Project Scope & Limitations
19/10/2026	M	Nguyen Chi Tung Duong	Standardize Acronyms, Project Purpose, Section Summary (GAP-04)
21/10/2026	M	Nguyen Chi tung Duong	Proposed Solution, Major Features (FE-01 to FE-18), and Limitations (LI-01 to LI-10) to align with Report 3 (SRS)

*A - Added M - Modified D - Deleted

Definition and Acronyms

Acronym	Definition
AI	Artificial Intelligence
GPS	Global Positioning System
QR	Quick Response (Code)
API	Application Programming Interface
GAP	Identified gap in existing solutions
FE	Feature
LI	Limitation / Exclusion
EPR	Extended Producer Responsibility
B2C	Business-to-Consumer
B2B	Business-to-Business
MVP	Minimum Viable Product
UI/UX	User Interface / User Experience
OTP	One-Time Password
SMTP	Simple Mail Transfer Protocol (Gmail SMTP)
FCM	Firebase Cloud Messaging
OAuth	Open Authorization (Google OAuth 2.0)
VietQR	Vietnam National Payment QR Standard (NAPAS 247)
RBAC	Role-Based Access Control
OSM	OpenStreetMap

I. Project Overview

1. Project Information

- English name: ***Intelligent Recyclable Waste Classification and Trade Matching System.***
- Vietnamese name: ***Hệ thống hỗ trợ phân loại và kết nối thu mua rác thải tái chế thông minh.***
- Project code: ***EcoLink***
- Group name: ***SEP490-G141***
- Software type: ***Web Application***

2. Project Purpose

The primary purpose of this project is to build a system that helps users easily identify types of recyclable waste through captured or uploaded images.

The system not only recognizes the type of waste but also provides localized, step-by-step sorting and pre-treatment instructions from a configurable Rule Engine so users know exactly how to handle each item.

Additionally, the project connects household users with verified local scrap collectors and recycling facilities through location-based discovery, direct handover requests, real-time chatbox negotiation, digital handover receipts, and transparent partner ratings, while sustaining long-term green living habits through gamification (Green Points, Virtual Tree, and reward redemption).

3. Project Stakeholders

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II. Product Background

Based on the practical issues outlined in the project's research, sorting waste at the source currently faces a massive barrier stemming from a very simple reality: users are often unsure where a specific type of waste should be thrown.

Imagine a daily scenario: You have just finished a meal in a styrofoam box or drank a bottle of water. Standing in front of trash bins labeled "Organic", "Recyclable", and "Other", you start to wonder. Should a styrofoam box with food residue go into the recycling bin or the other bin? For a plastic bottle, which bin does it go to? Or for items like old batteries, how can they be handled since they cannot be thrown out with regular household waste?

Because there are no ready answers at that moment, users still have to make their own decisions on the spot. Even with a good intention to protect the environment, this confusion leads to improper disposal, causing recyclable materials to be ruined or sent to landfills.

The core problem here is two-fold. First, when looking at an object, users simply do not know what category it belongs to (e.g., whether it is plastic, paper, metal, or other waste). Second, even if they know what the item is (for example, a plastic bottle), they do not know the proper handling process, such as whether it needs to be washed, if it is recyclable, or which specific bin it belongs to.

This is the exact reason the EcoLink system was created. Instead of forcing users to memorize complicated waste sorting rules, the system provides an extremely simple workflow: the user takes or uploads a photo of the waste, the AI recognizes the item, and the system looks up the sorting rules to provide a clear instruction. For example, if a user snaps a picture of a water bottle, the system will immediately identify it as a plastic bottle and provide visual handling instructions, such as: "1. Empty liquid, 2. Rinse if necessary, 3. Put into recycling bin". By providing guidance right at the moment of disposal, the product turns a confusing daily chore into a confident and eco-friendly habit.

III. Existing Solutions

Four categories of solutions currently address parts of the recyclable waste problem. The analysis below evaluates each against the specific context of EcoLink: a Vietnamese platform that must (1)

identify waste from a photo, (2) deliver localised sorting instructions, and (3) connect users with trade partners who will actually purchase the material.

1. Recycle Coach

Recycle Coach is a waste-management mobile application developed by Recycle Coach Inc. (Canada). It is deployed in partnership with hundreds of municipalities across North America and Australia, allowing each city to configure its own rules and collection schedules directly inside the platform.

Website / Link	https://www.recyclecoach.com/
Target Users	Urban and suburban residents in North America, Australia; municipalities seeking a citizen-facing waste tool
Core Features	"What Goes Where?" text-based item search; AI-assisted camera scan (limited rollout); personalised curbside collection calendar; automated pickup reminders; drop-off depot locator; in-app issue reporting to local authorities
Strengths	Comprehensive rule database tied to real municipal policies; official integration with local governments ensures accuracy; calendar and reminder features drive repeat engagement
Weaknesses	Camera recognition is optional and inconsistently available across municipalities; the platform has no trade or purchase mechanism — it only tells users *where* to drop items, not *who will buy* them; service coverage is concentrated in North America and Australia
GAP for this project	Recycle Coach is built around the assumption that a municipality will manage the platform and keep its database current. In Vietnam, no such municipality-level digital infrastructure exists. Furthermore, the app has no mechanism for users to connect with private scrap-buying partners (thu mua) or to complete a buy-sell transaction. A Vietnamese user who correctly identifies a plastic bottle using Recycle Coach is still left with no actionable next step for monetising or handing over that material.

2. iRecycle

iRecycle, developed by Earth911 (USA), was one of the largest recycling-locator tools in North America, providing access to a database of over 100,000 recycling and disposal facilities across hundreds of material categories.

Website / Link	https://search.earth911.com/
Target Users	US-based residents seeking drop-off or recycling locations for specific materials
Core Features	Location and ZIP-code-based search for nearby recycling facilities; category browsing (plastics, metals, electronics, batteries, etc.); facility details including contact, hours, and accepted materials
Strengths	Extensive material database covering rare items such as motor oil, paint, and e-waste; the most comprehensive recycling-location tool available for the US market
Weaknesses	Geographic coverage is almost exclusively the United States; no AI image recognition — users must already know what material they have; no buying/selling functionality; mobile app has been discontinued, leaving only a web tool; no personalisation or gamification
GAP for this project	iRecycle is a static directory, not an intelligent assistant. It requires users to know the exact material name before searching — precisely the knowledge gap that EcoLink targets. Beyond geography (the tool has zero Vietnam coverage), iRecycle cannot help a user who holds an ambiguous object and wonders

	whether it is recyclable. It also has no mechanism for a private recycling trader to publish their buying price, availability, or accepted categories, which is the second core value EcoLink must deliver.

3. Litterati

Litterati is a US-based citizen-science application that uses AI (branded as LitterAI) to classify litter items photographed during community clean-up activities. Its primary goal is to generate structured environmental datasets that NGOs and policymakers can use to drive systemic change.

Website / Link	https://www.litterati.org/
Target Users	Environmental volunteers, NGOs, citizen-science communities, schools, corporate CSR programs
Core Features	Photo capture with AI-assisted tagging (Object–Material–Brand taxonomy); GPS geotagging and timestamping; community litter maps; brand-pollution analytics; data export for policymakers
Strengths	The only solution reviewed that applies AI image analysis directly to waste photos; large-scale dataset validated by National Science Foundation funding; effective at creating accountability for brand-level plastic pollution
Weaknesses	The AI classifies litter for *research and accountability purposes*, not for giving individual users sorting instructions; the app is designed for outdoor litter photography, not for household disposal decisions; there is no knowledge base of sorting rules, no handling guidance, and no connection to recycling buyers
GAP for this project	Litterati demonstrates that AI-based waste photo classification is feasible, but it is built entirely for community data collection — not for the moment-of-disposal decision a household user faces. It answers "what brand polluted here?" rather than "what should I do with this item right now?". Litterati does not provide step-by-step handling instructions, has no knowledge base tied to Vietnamese waste policy, and has no feature for connecting users with trade partners. It addresses the analytical layer while leaving the practical user need unsolved.

4. VECA

VECA is a Vietnamese technology startup (founded 2021) that built an "Uber-like" mobile platform to connect households wanting to sell recyclable scrap with licensed scrap collectors and recycling facilities. It was the closest local equivalent to EcoLink's trade-matching feature.

Website / Link	https://veca.app/doanh-nghiep/phe-lieu-cong-nghiep
Target Users	Urban Vietnamese households with recyclable scrap to sell; scrap collectors and recycling businesses (B2B focus since late 2024)
Core Features	On-demand home scrap-collection booking; transparent price list for common materials (plastic, paper, metal); digital management tools for scrap yards (VECA Vựa Số); e-wallet payment integration; corporate partnership program (EPR compliance)
Strengths	First mover in digitising Vietnam's traditionally informal ve-chai (scrap collector) sector; transparent, standardised pricing reduces exploitation of both sellers and collectors; praised for intuitive UI despite a technically diverse user base
Weaknesses	B2C home-collection service was officially discontinued on 1 October 2024 due to unsustainable unit economics (small-volume orders were unprofitable); no AI waste-recognition feature — users must already know what they have and how much of it; no sorting guidance or knowledge base; pivoted fully to B2B

	scrap-yard management tools, leaving individual household users without a solution
GAP for this project	VECA proved that Vietnamese users are willing to engage with a digital recycling platform when price transparency and convenience are offered. However, the discontinuation of its B2C service exposed the unsolved problem: individual users produce small, mixed, and often unidentified waste that does not fit neatly into a booking-based model. EcoLink addresses the layer that VECA skipped — helping users first identify and classify what they have — before connecting them to a trade partner. The addition of AI recognition, a sorting knowledge base, and a gamified engagement loop is precisely what is needed to make the connection sustainable at the individual-user level.

5. Section Summary

Each of the solutions reviewed addresses at most one of EcoLink's core requirements. Recycle Coach and iRecycle provide sorting guidance but are unavailable in Vietnam, rely on text-based input (requiring users to already know what they hold), and have no mechanism for completing a trade transaction. Litterati introduces AI image classification but uses it exclusively for environmental research, providing no actionable disposal instructions and no buyer connection. VECA demonstrated real demand for a Vietnamese recycling trade platform but folded its B2C service because it never solved the upstream problem: users did not know what waste they had or how to handle it, nor did it offer a lightweight engagement mechanism to retain small-volume household contributors. No existing solution combines AI-powered visual identification, localized Vietnamese sorting guidance, a verified recycler discovery and handover coordination workflow, and a gamified retention loop into a single platform — which is the specific gap EcoLink is designed to fill.

These unresolved deficiencies define four distinct gaps that EcoLink must address: GAP-01 (Waste Identification Gap) — no solution provides AI-powered photo-based identification accessible to Vietnamese users with continuous feedback curation; GAP-02 (Sorting & Handling Guidance Gap) — no solution delivers localized, rule-based disposal instructions and educational content at the moment of need; GAP-03 (Recycling Trade Connectivity Gap) — no solution connects individual households with verified local buyers through GPS discovery, direct handover requests, in-app negotiation, and confirmed receipts; and GAP-04 (User Engagement & Retention Gap) — no solution provides gamified incentives (Green Points, Virtual Tree progression, and reward redemption) to sustain daily household sorting and recycling participation.

IV. Proposed Solution

The platform is an intelligent waste sorting and recycling trade coordination solution designed to bridge the critical gap between environmental intent and daily execution. As identified in the gap analysis above, no existing solution simultaneously addresses waste identification, localized sorting guidance, verified recycler connectivity, and user retention in the Vietnamese context: international tools such as Recycle Coach and iRecycle lack both local coverage and any trade mechanism, Litterati's AI serves research rather than individual disposal decisions, and VECA's discontinued B2C service demonstrated that connecting users with buyers fails without first solving the upstream identification and engagement problem. Instead of leaving consumers to second-guess static bin labels or misclassify ambiguous household items, EcoLink immediately clarifies sorting decisions by identifying waste items from everyday snapshots and providing direct disposal instructions. Crucially,

the platform closes the recycling lifecycle loop by eliminating market friction: it enables households holding accumulated recyclables to discover nearby verified scrap collectors and recycling facilities via GPS, send direct handover requests, negotiate schedules and terms via an in-app chatbox, and confirm digital handover receipts (with cash or bank settlement handled directly between parties outside the system).

Two core stakeholders benefit directly from this solution. For everyday consumers and households, the immediate gain is the complete removal of decision fatigue at the bin, transforming an inconvenient chore into a frictionless habit while creating tangible financial and gamified incentives by turning segregated scrap into fair cash returns and redeemable Green Points. For scrap collectors and recycling facilities, the platform provides a verified online presence with transparent purchasing prices, a steady stream of pre-sorted, low-contamination recyclables through direct handover requests containing photos and estimated weights, and structured in-app negotiation and appointment scheduling that reduce manual sorting overhead and price disputes.

A photo-based identification approach coupled with direct recycler discovery, handover coordination, and gamified rewards is the right strategy today because municipal mandates for waste segregation are tightening, yet compliance consistently falters at the moment of disposal. The proven demand for a Vietnamese trade platform — evidenced by VECA's early user adoption before its B2C service became economically unsustainable — confirms that the market incentive exists; what was missing was the upstream layer that helps users understand and prepare their waste before a handover begins, alongside a gamified engagement loop (daily check-ins, Virtual Tree growth, and vouchers) that keeps users active between transactions.

V. Project Scope & Limitations

GAP-01: Waste Identification Gap
GAP-02: Sorting & Handling Guidance Gap
GAP-03: Recycling Trade Connectivity Gap
GAP-04: User Engagement & Retention Gap

1. Major Features

ID	Feature Name	Description	Addressed Gaps
FE-01	AI-Based Waste Identification & Classification	Users capture or upload a photo of a waste item, and the trained AI Vision model (YOLOv8) analyzes the image to predict its recyclable waste category, confidence score, and estimated scrap price range. Users confirm or adjust the predicted category before the scan is saved to history (U-01).	GAP-01
FE-02	Localized Waste Sorting Guidance & Catalog Search	Based on an AI scan confirmation or manual keyword/material filter search in the Waste Catalog, the system retrieves step-by-step cleaning, pre-treatment, and bin disposal instructions together with recyclability status and reference pricing from the Rule Engine (U-02, U-03).	GAP-02

FE-03	Waste Knowledge Base & Sorting Policy Management	Administrators define, update, or deprecate master waste categories and configure localized sorting rules and handling steps within the Rule Engine so that guidance stays accurate without retraining the AI model (A-05, A-06).	GAP-01, GAP-02
FE-04	Recycler Profile, Pricing & Availability Management	Approved scrap collectors and recycling facilities maintain their accepted waste categories, reference prices (VND/kg), minimum collection weights, and operating schedules, and can toggle their real-time Online/Offline availability on the public GPS map (R-01, R-02).	GAP-03
FE-05	Recycler Onboarding & Verification	Authenticated citizens can submit an upgrade application with facility details, GPS coordinates, accepted categories, and business/identity documents to become a Recycler partner. Operations Staff review and verify applications before granting the Recycler role (U-17, S-04).	GAP-03
FE-06	Location-Based Recycler Discovery	Users search for active, approved scrap collectors and recycling facilities on an interactive map within a configurable GPS radius (2–20 km) using PostGIS spatial queries, filtered by accepted waste category and sorted by ascending distance (U-05).	GAP-03
FE-07	Handover Request & Appointment Coordination	Users submit direct handover requests to a selected Recycler specifying waste categories, estimated weights, photos, and a preferred collection time slot, and can update or cancel pending requests. Recyclers review incoming requests to accept (confirming the appointment) or reject them, or cancel scheduled appointments with a recorded reason (U-06, U-07, R-03, R-04).	GAP-03
FE-08	In-App Negotiation Chatbox	Users and Recyclers exchange real-time messages, photo attachments, and structured price/quantity/time counter-offers within a handover request to finalize collection details prior to completion (U-08, R-05).	GAP-03
FE-09	Handover Receipt & Transaction Confirmation	Upon on-site collection, Recyclers record actual weighed quantities, upload scale proof photos, and issue a digital handover receipt. Users review and confirm the receipt after settling cash/bank payment directly outside the system, completing the transaction and earning Green Points, and can view their transaction history (R-06, U-09, U-12).	GAP-03
FE-10	Partner Rating, Review & Complaint Resolution	Users submit 1–5 star ratings and reviews for Recyclers after completed handovers, or lodge formal transaction complaints with photo evidence when weight or settlement discrepancies occur. Operations Staff investigate chat logs and receipts to issue binding dispute resolutions (U-10, U-11, S-05).	GAP-03
FE-11	Multi-Channel Notifications	The system delivers in-app notification center alerts, Firebase Cloud Messaging (FCM) web push notifications, and automated transactional emails via Gmail SMTP for account verification, handover updates, chat messages, dispute outcomes, and point awards (SU-04).	GAP-03
FE-12	User Authentication & Profile Management	Visitors and registered users can browse the public homepage, register via email OTP/activation link or Google OAuth 2.0 Single Sign-On, sign in/out, recover passwords via email, change passwords, and update personal profiles and avatars (G-01, G-02, SU-01 to SU-03, SU-05, SU-06).	Supporting (Platform Core)

FE-13	Gamification, Virtual Tree & Rewards	Users earn Green Points from completed handovers and daily check-ins, redeem points for gifts/vouchers in the Gift Store, and water their Virtual Tree to unlock environmental Eco Facts. Staff manage Eco Facts and gift inventory, while Administrators configure point conversion rules and leaderboard cycles (U-13 to U-15, S-09, S-10, A-07, A-08).	GAP-04
FE-14	In-System Service Packages via VietQR	Users and Recyclers can purchase internal platform service packages (e.g., Ad-Free experience, Verified Partner badge, priority map placement) via dynamic VietQR codes. Administrators configure package fees and perform monthly financial reconciliation of service revenue (U-16, R-07, A-10, A-11).	Supporting (Platform Operations)
FE-15	AI Misclassification Feedback & Dataset Curation	Users report incorrect AI predictions with suggested true labels. Operations Staff review reported images, apply standardized annotations, and curate verified images into versioned datasets for future YOLOv8 model retraining (U-04, S-01 to S-03).	GAP-01
FE-16	Environmental Blog & Content Management	Operations Staff draft educational articles and green-living blog posts, and Administrators review, approve, or reject submissions before public publication (S-08, A-09).	GAP-02
FE-17	Account, Staff & Permission Governance	Operations Staff view User/Recycler accounts, lock or unlock violating accounts with audit logging, and record daily shift check-ins. Administrators manage Staff accounts and configure granular Role-Based Access Control (RBAC) permissions (S-06, S-07, S-11, A-03, A-04).	Supporting (Governance)
FE-18	System Analytics Dashboard & Report Export	Administrators monitor platform KPIs (recycled tonnage, AI accuracy & misclassification rates, active users, service revenue) on an executive dashboard and export operational and financial reports to Excel/CSV (A-01, A-02).	Supporting (Governance)

2. Limitations & Exclusions

ID	Excluded Item	Reason
LI-01	Native Mobile Application	The current version is developed as a responsive web application. Dedicated Android and iOS native applications require additional development, testing, and maintenance effort and are outside the scope of the initial capstone release.
LI-02	Automated Physical Waste Sorting	EcoLink provides software-based waste identification and sorting guidance only. Integration with robotic sorting systems, smart bins, conveyor systems, or other physical waste-processing equipment is not included in this version.
LI-03	Recognition of Unsupported Waste Categories	The AI model is limited to the 8–12 recyclable waste categories represented in the project's training dataset. Objects or waste types outside the supported classes are not guaranteed to be identified correctly.
LI-04	Automatic Waste Weight and Quality Verification	The web application cannot physically verify the exact weight, cleanliness, contamination level, or material quality of recyclable waste. Such information is estimated by the user and verified on-site by the recycler via scale proof photos in the digital handover receipt.

LI-05	Waste Transportation & Fleet Management	EcoLink connects users with scrap collectors and recycling facilities but does not directly operate transportation services or manage vehicles, drivers, fleet schedules, or physical waste collection operations.
LI-06	Nationwide Recycler Coverage Guarantee	Recycler discovery depends on the approved collectors and recycling facilities registered and active in the EcoLink database. The system therefore cannot guarantee complete coverage of all recycling purchasers throughout Vietnam.
LI-07	Guaranteed AI Classification Accuracy	Classification accuracy depends on factors such as the training dataset, image quality, lighting conditions, viewing angle, background, and similarity between waste categories. AI results are intended to assist users and require user confirmation before recording.
LI-08	Real-Time Recycling Market Price Prediction	EcoLink displays reference purchasing prices (VND/kg) configured by recyclers and allows price negotiation via chatbox, but it does not operate as a commodity exchange and does not predict or guarantee real-time market prices for recyclable materials.
LI-09	Guaranteed Trade Completion	An accepted handover request only confirms the scheduled appointment and initial terms between user and recycler. EcoLink cannot guarantee that the physical collection, on-site weighing, or direct settlement between the two parties will be successfully completed.
LI-10	In-System Settlement of Waste Trade Transactions	Cash or bank payments for buying and selling recyclable waste are settled directly between the User and the Recycler outside the platform at the time of physical handover. EcoLink only records confirmed receipt figures and does not process, hold in escrow, or disburse trade payout funds; dynamic VietQR integration is strictly limited to purchasing internal platform service packages.